

Excerpts from Prof. Girishkumar's report submitted to secretary, DoT, on cellphone radiation effects in Dec. 2010

1. Effect of increased dose of radiation on sleep

- (i) At 0.1 nW/cm² exposure, sleep disturbance is 25 per cent
 - (ii) At 1 nW/cm², sleep disturbance is 35 per cent
 - (iii) At 10 nW/cm², sleep disturbance is 45 per cent
 - (iv) At 100 nW/cm², sleep disturbance is 55 per cent
- It can be seen that with logarithmic increase in dose, sleep disturbance increases linearly.

2. Radiation in Mumbai

A recent study revealed that the electromagnetic radiations emitted from phone towers in Mumbai were way beyond safety levels. According to the study, the real safe limit of radiations for human beings is up to 50 $\mu\text{W}/\text{m}^2$ and the upper limit is 100 $\mu\text{W}/\text{m}^2$. The study revealed that the radiation readings in several residential areas were more than 1000 $\mu\text{W}/\text{m}^2$. Mumbai has witnessed a spurt in the number of phone towers in the recent past exposing people to dangerously high levels of radiations. Being exposed to such radiation leads to problems like loss of appetite, irritation, fatigue, difficulty in concentrating, headaches, problems in vision, infertility and deformity of foetus.

3. Occupational hazard

Studies of people who are exposed to mobile radiations as part of their work have shown elevated levels of health risks in them. Those employed in electrical occupations are 10 to 20 per cent more likely to get brain cancer and show elevations in non-Hodgkin's lymphoma.

4. Auditory perception of microwaves

Ringxiety—the psychological disease of hearing phantom sound and sensation of cellphone ring—has appeared in Japan (also reported in India) among cellphone users.

research papers and conclusions arrived at

- 4. To look for proven research
- 5. To study the research carried out in India
- 6. To identify the power level suited for India, which is a tropical country

- 7. To give suggestions to the users to avoid problems
- 8. To give suggestions to the government regarding future actions
- 9. To give suggestions for immediate implementation
- 10. To go through case studies in India and abroad

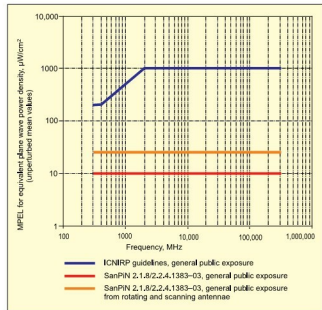


Fig. 4: Comparison of MPEL at 0.03-300GHz power flux density established as the obligatory standard for the general public of Russia versus similar values recommended by ICNIRP

mobile service providers. Whether this level of power reduction is adequate or not is the next question because many countries have already kept power levels much lower than this level in mobile transmitters.

Thus, in the middle of projections of epidemiological proportions of health issues, this topic gains much importance. Even more important is the point that regulations followed in other countries are not followed by mobile service providers in India even to maintain the high-level ITU specifications.

Recommendations to the government

1. Immediately implement the 50-metre safety zone around mobile towers.
2. Separate out different service providers' mobile towers, otherwise, these tend to accumulate more power.
3. Avoid towers near educational institutions.
4. Ensure that SAR rating is provided on the front cover of the cellphone manual with warning.
5. Implement EMF monitoring systems near towers and other locations.
6. Ensure that the ICMR research set-up concentrates on areas where cellphones were introduced first in India and speed up the process.
7. Tie up with countries having rich experience to evaluate the long-term effect of cellphone usage, as in other countries cellphones were introduced much earlier.
8. Ensure that companies supply necessary accessories to mobile users as a preventive action and advertise the harmful effects of cellphones strongly to the public.

Cautions for cellphone users

Mobile companies should accept the fact that radiations cause serious health problems. In the last few years, epidemiology evidence on the mobile phone use and risk of brain and other tumours of the head has

- 11. To list out possible health effects in India
- 12. To analyse the effects on different age groups

An inter-ministerial committee has been set up, which has given its recommendations on the subject. The committee's recommendations have far-reaching implications as it has suggested to cut the mobile power transmitted by one-tenth of its current value. This has serious implications for the